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# You need to install these libraries first:
# pip install torch transformers datasets

import torch
from transformers import AutoTokenizer, AutoModelForCausalLM, TrainingArguments,
Trainer, pipeline, DataCollatorForLanguageModeling
from datasets import Dataset

# 1. Prepare your custom data
# The format should be a list of strings, where each string represents a prompt
# and its desired completion. A simple Q&A format works well.
all_data = [
    "What is the capital of Afghanistan? Kabul is the capital of Afghanistan.",
    "What is the capital of Albania? Tirana is the capital of Albania.",
    "What is the capital of Algeria? Algiers is the capital of Algeria.",
    "What is the capital of Andorra? Andorra la Vella is the capital of Andorra.",
    "What is the capital of Angola? Luanda is the capital of Angola.",
    "What is the capital of Antigua and Barbuda? Saint John's is the capital of Antigua
and Barbuda.",
    "What is the capital of Argentina? Buenos Aires is the capital of Argentina.",
    "What is the capital of Armenia? Yerevan is the capital of Armenia.",
    "What is the capital of Australia? Canberra is the capital of Australia.",
    "What is the capital of Austria? Vienna is the capital of Austria.",
    "What is the capital of Azerbaijan? Baku is the capital of Azerbaijan.",
    "What is the capital of Bahamas? Nassau is the capital of Bahamas.",
    "What is the capital of Bahrain? Manama is the capital of Bahrain.",
    "What is the capital of Bangladesh? Dhaka is the capital of Bangladesh.",
    "What is the capital of Barbados? Bridgetown is the capital of Barbados.",
    "What is the capital of Belarus? Minsk is the capital of Belarus.",
    "What is the capital of Belgium? Brussels is the capital of Belgium.",
    "What is the capital of Belize? Belmopan is the capital of Belize.",
    "What is the capital of Benin? Porto-Novo is the capital of Benin.",
    "What is the capital of Bhutan? Thimphu is the capital of Bhutan.",
    "What is the capital of Bolivia? Sucre is the capital of Bolivia.",
    "What is the capital of Bosnia and Herzegovina? Sarajevo is the capital of Bosnia and
Herzegovina.",
    "What is the capital of Botswana? Gaborone is the capital of Botswana.",
    "What is the capital of Brazil? Brasília is the capital of Brazil.",
    "What is the capital of Brunei? Bandar Seri Begawan is the capital of Brunei.",
    "What is the capital of Bulgaria? Sofia is the capital of Bulgaria.",
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"What is the capital of Burkina Faso? Ouagadougou is the capital of Burkina Faso.",
"What is the capital of Burundi? Gitega is the capital of Burundi.",
"What is the capital of Cabo Verde? Praia is the capital of Cabo Verde.",
"What is the capital of Cambodia? Phnom Penh is the capital of Cambodia.",
"What is the capital of Cameroon? Yaoundé is the capital of Cameroon.",
"What is the capital of Canada? Ottawa is the capital of Canada.",
"What is the capital of Central African Republic? Bangui is the capital of Central African Republic.",
"What is the capital of Chad? N'Djamena is the capital of Chad.",
"What is the capital of Chile? Santiago is the capital of Chile.",
"What is the capital of China? Beijing is the capital of China.",
"What is the capital of Colombia? Bogotá is the capital of Colombia.",
"What is the capital of Comoros? Moroni is the capital of Comoros.",
"What is the capital of Costa Rica? San José is the capital of Costa Rica.",
"What is the capital of Croatia? Zagreb is the capital of Croatia.",
"What is the capital of Cuba? Havana is the capital of Cuba.",
"What is the capital of Cyprus? Nicosia is the capital of Cyprus.",
"What is the capital of Czech Republic? Prague is the capital of Czech Republic.",
"What is the capital of Denmark? Copenhagen is the capital of Denmark.",
"What is the capital of Djibouti? Djibouti is the capital of Djibouti.",
"What is the capital of Dominica? Roseau is the capital of Dominica.",
"What is the capital of Dominican Republic? Santo Domingo is the capital of Dominican Republic.",
"What is the capital of East Timor? Dili is the capital of East Timor.",
"What is the capital of Ecuador? Quito is the capital of Ecuador.",
"What is the capital of Egypt? Cairo is the capital of Egypt.",
"What is the capital of El Salvador? San Salvador is the capital of El Salvador.",
"What is the capital of Equatorial Guinea? Malabo is the capital of Equatorial Guinea.",
"What is the capital of Eritrea? Asmara is the capital of Eritrea.",
"What is the capital of Estonia? Tallinn is the capital of Estonia.",
"What is the capital of Eswatini? Mbabane is the capital of Eswatini.",
"What is the capital of Ethiopia? Addis Ababa is the capital of Ethiopia.",
"What is the capital of Fiji? Suva is the capital of Fiji.",
"What is the capital of Finland? Helsinki is the capital of Finland.",
"What is the capital of France? Paris is the capital of France.",
"What is the capital of Gabon? Libreville is the capital of Gabon.",
"What is the capital of Gambia? Banjul is the capital of Gambia.",
"What is the capital of Georgia? Tbilisi is the capital of Georgia.",
"What is the capital of Germany? Berlin is the capital of Germany.",
"What is the capital of Ghana? Accra is the capital of Ghana.",
"What is the capital of Greece? Athens is the capital of Greece.",

"What is the capital of Grenada? St. George's is the capital of Grenada.",
"What is the capital of Guatemala? Guatemala City is the capital of Guatemala.",
"What is the capital of Guinea? Conakry is the capital of Guinea.",
"What is the capital of Guinea-Bissau? Bissau is the capital of Guinea-Bissau.",
"What is the capital of Guyana? Georgetown is the capital of Guyana.",
"What is the capital of Haiti? Port-au-Prince is the capital of Haiti.",
"What is the capital of Honduras? Tegucigalpa is the capital of Honduras.",
"What is the capital of Hungary? Budapest is the capital of Hungary.",
"What is the capital of Iceland? Reykjavik is the capital of Iceland.",
"What is the capital of India? New Delhi is the capital of India.",
"What is the capital of Indonesia? Jakarta is the capital of Indonesia.",
"What is the capital of Iran? Tehran is the capital of Iran.",
"What is the capital of Iraq? Baghdad is the capital of Iraq.",
"What is the capital of Ireland? Dublin is the capital of Ireland.",
"What is the capital of Israel? Jerusalem is the capital of Israel.",
"What is the capital of Italy? Rome is the capital of Italy.",
"What is the capital of Ivory Coast? Yamoussoukro is the capital of Ivory Coast.",
"What is the capital of Jamaica? Kingston is the capital of Jamaica.",
"What is the capital of Japan? Tokyo is the capital of Japan.",
"What is the capital of Jordan? Amman is the capital of Jordan.",
"What is the capital of Kazakhstan? Nur-Sultan is the capital of Kazakhstan.",
"What is the capital of Kenya? Nairobi is the capital of Kenya.",
"What is the capital of Kiribati? Tarawa is the capital of Kiribati.",
"What is the capital of Korea, North? Pyongyang is the capital of Korea, North.",
"What is the capital of Korea, South? Seoul is the capital of Korea, South.",
"What is the capital of Kuwait? Kuwait City is the capital of Kuwait.",
"What is the capital of Kyrgyzstan? Bishkek is the capital of Kyrgyzstan.",
"What is the capital of Laos? Vientiane is the capital of Laos.",
"What is the capital of Latvia? Riga is the capital of Latvia.",
"What is the capital of Lebanon? Beirut is the capital of Lebanon.",
"What is the capital of Lesotho? Maseru is the capital of Lesotho.",
"What is the capital of Liberia? Monrovia is the capital of Liberia.",
"What is the capital of Libya? Tripoli is the capital of Libya.",
"What is the capital of Liechtenstein? Vaduz is the capital of Liechtenstein.",
"What is the capital of Lithuania? Vilnius is the capital of Lithuania.",
"What is the capital of Luxembourg? Luxembourg City is the capital of Luxembourg.",
"What is the capital of Madagascar? Antananarivo is the capital of Madagascar.",
"What is the capital of Malawi? Lilongwe is the capital of Malawi.",
"What is the capital of Malaysia? Kuala Lumpur is the capital of Malaysia.",
"What is the capital of Maldives? Malé is the capital of Maldives.",
"What is the capital of Mali? Bamako is the capital of Mali.",
"What is the capital of Malta? Valletta is the capital of Malta.",

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"What is the capital of Marshall Islands? Majuro is the capital of Marshall Islands.",
"What is the capital of Mauritania? Nouakchott is the capital of Mauritania.",
"What is the capital of Mauritius? Port Louis is the capital of Mauritius.",
"What is the capital of Mexico? Mexico City is the capital of Mexico.",
"What is the capital of Micronesia? Palikir is the capital of Micronesia.",
"What is the capital of Moldova? Chişinău is the capital of Moldova.",
"What is the capital of Monaco? Monaco is the capital of Monaco.",
"What is the capital of Mongolia? Ulaanbaatar is the capital of Mongolia.",
"What is the capital of Montenegro? Podgorica is the capital of Montenegro.",
"What is the capital of Morocco? Rabat is the capital of Morocco.",
"What is the capital of Mozambique? Maputo is the capital of Mozambique.",
"What is the capital of Myanmar? Naypyidaw is the capital of Myanmar.",
"What is the capital of Namibia? Windhoek is the capital of Namibia.",
"What is the capital of Nauru? Yaren is the capital of Nauru.",
"What is the capital of Nepal? Kathmandu is the capital of Nepal.",
"What is the capital of Netherlands? Amsterdam is the capital of Netherlands.",
"What is the capital of New Zealand? Wellington is the capital of New Zealand.",
"What is the capital of Nicaragua? Managua is the capital of Nicaragua.",
"What is the capital of Niger? Niamey is the capital of Niger.",
"What is the capital of Nigeria? Abuja is the capital of Nigeria.",
"What is the capital of North Macedonia? Skopje is the capital of North Macedonia.",
"What is the capital of Norway? Oslo is the capital of Norway.",
"What is the capital of Oman? Muscat is the capital of Oman.",
"What is the capital of Pakistan? Islamabad is the capital of Pakistan.",
"What is the capital of Palau? Ngerulmud is the capital of Palau.",
"What is the capital of Panama? Panama City is the capital of Panama.",
"What is the capital of Papua New Guinea? Port Moresby is the capital of Papua New Guinea.",
"What is the capital of Paraguay? Asunción is the capital of Paraguay.",
"What is the capital of Peru? Lima is the capital of Peru.",
"What is the capital of Philippines? Manila is the capital of Philippines.",
"What is the capital of Poland? Warsaw is the capital of Poland.",
"What is the capital of Portugal? Lisbon is the capital of Portugal.",
"What is the capital of Qatar? Doha is the capital of Qatar.",
"What is the capital of Romania? Bucharest is the capital of Romania.",
"What is the capital of Russia? Moscow is the capital of Russia.",
"What is the capital of Rwanda? Kigali is the capital of Rwanda.",

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# 2. Split data into training and validation sets
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# We'll use the first 140 entries for training and the last 2 for validation.
train_data = all_data[:142]
eval_data = all_data[10:]

# 3. Convert your data to a Hugging Face Dataset format
# We'll create separate datasets for training and evaluation.
train_dataset = Dataset.from_dict({'text': train_data})
eval_dataset = Dataset.from_dict({'text': eval_data})

# 4. Load the pre-trained model and tokenizer
model_name = "distilgpt2"
tokenizer = AutoTokenizer.from_pretrained(model_name)
model = AutoModelForCausalLM.from_pretrained(model_name)

# Set the pad token. Some models (like GPT-2) don't have one by default,
# but it's needed for training. We'll set it to the end-of-sentence token.
tokenizer.pad_token = tokenizer.eos_token

# 5. Tokenize the datasets
# This function prepares the data for the model by converting text to token IDs.
def tokenize_function(examples):
    # We use padding='max_length' and truncation=True to make sure all sequences
    # have the same length and fit within the model's context window.
    return tokenizer(examples['text'], padding="max_length", truncation=True,
max_length=128)

# We use .map() to apply the tokenization function to the entire dataset.
tokenized_train_dataset = train_dataset.map(tokenize_function, batched=True)
tokenized_eval_dataset = eval_dataset.map(tokenize_function, batched=True)

# 6. Set up the Training Arguments
# We add the evaluation strategy to also evaluate on the validation set.
training_args = TrainingArguments(
    output_dir="/Users/srinip/finetuned2", # Changed to your desired directory
    num_train_epochs=5,
    per_device_train_batch_size=2,
    learning_rate=5e-5,
    save_strategy="epoch",
    eval_steps=100, # Added a step count for evaluation
)

# 7. Initialize the Trainer

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# The Trainer now receives both the training and evaluation datasets.
# We also add the data_collator.
data_collator = DataCollatorForLanguageModeling(tokenizer=tokenizer, mlm=False)

trainer = Trainer(
    model=model,
    args=training_args,
    train_dataset=tokenized_train_dataset,
    eval_dataset=tokenized_eval_dataset, # Pass the evaluation dataset here
    data_collator=data_collator, # This will automatically create the labels
    tokenizer=tokenizer,
)

# 8. Start the training and evaluation process
print("Starting fine-tuning...")
trainer.train()
print("Fine-tuning complete. Model saved to /Users/srinip/finetuned2/")

print("\nValidation complete.")

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import gradio as gr
from transformers import pipeline, AutoTokenizer, AutoModelForCausalLM

# 1. Initialize the Text Generation pipeline with the fine-tuned model
# The model will be loaded from the local directory created by the fine-tuning script.
print("Loading fine-tuned Text Generation model and pipeline...")

# Define the path to your fine-tuned model.
# The `from_pretrained` method will automatically find the model and tokenizer files.
model_path = "/Users/srinip/finetuned2/checkpoint-355"

try:
    # Load the model and tokenizer from the saved directory.
    finetuned_model = AutoModelForCausalLM.from_pretrained(model_path)
    finetuned_tokenizer = AutoTokenizer.from_pretrained(model_path)

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    # Create a text generation pipeline with the fine-tuned model and tokenizer.
    qa_pipeline = pipeline("text-generation", model=finetuned_model,
tokenizer=finetuned_tokenizer)
    print("Pipeline created successfully!")
except Exception as e:
    # Handle case where the model is not found.
    print(f"Error loading fine-tuned model: {e}")
    print("Please make sure you have run the fine-tuning script first.")
    # Exit or provide a placeholder pipeline to prevent the app from crashing.
    qa_pipeline = None

# 2. Define the function that Gradio will use to generate a response.
def answer_question(question):
    """
    Generates a text response based on a question using the fine-tuned
    Hugging Face text generation pipeline.

    Args:
        question (str): The question or prompt to be answered.

    Returns:
        str: The generated text response.
    """
    # Check if the pipeline was successfully loaded.
    if not qa_pipeline:
        return "Model not loaded. Please run the fine-tuning script first."

    # Ensure the question is provided.
    if not question:
        return "Please enter a question to get a response."

    # Use the pipeline to generate a response based on the question.
    # We set max_length and num_return_sequences to control the output.
    response = qa_pipeline(question, max_length=len(question.split()) + 20,
num_return_sequences=1)

    # The pipeline returns a list of dictionaries; we extract the generated text.
    generated_text = response[0]['generated_text']

    # The generated text will contain the original question. We'll clean this up
    # to return just the response.

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    if question in generated_text:
        start_index = generated_text.find(question) + len(question)
        final_answer = generated_text[start_index:].strip()
    else:
        final_answer = generated_text

    return final_answer

# 3. Create the Gradio Interface
# The interface now has a single input for the question and an output for the answer.
with gr.Blocks(theme=gr.themes.Monochrome()) as demo:
    gr.Markdown("# Simple Q&A Chatbot")
    gr.Markdown("This app uses a **fine-tuned** model to answer your questions about country capitals.")

    # Create a single textbox for the user's question.
    question_input = gr.Textbox(lines=5, label="Question", placeholder="e.g., What is the capital of Spain?")

    # Create the output textbox to display the answer.
    output_text = gr.Textbox(label="Answer")

    # Create the "Get Answer" button.
    submit_button = gr.Button("Get Answer")

    # Link the button to the function and the UI components.
    submit_button.click(
        fn=answer_question,
        inputs=question_input,
        outputs=output_text
    )

    # Example inputs to make it easy for the user to try the app.
    gr.Examples(
        examples=[
            ["What is the capital of Italy?"],
            ["What is the capital of France?"],
            ["What is the capital of Germany?"],
        ],
        inputs=question_input
    )

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# 4. Launch the Gradio app
if __name__ == "__main__":
    demo.launch()
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